

Quantum Mechanics III

Homework 12

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Topic

Control and Error Channels

Submission

Submit one PDF. Show the derivation, identify every approximation, and include a short consistency check. The maximum file size is 25 MB.

Problems

1. Prove the first displayed relation used in Control and Error Channels. State all assumptions and conventions.

$$U = T \exp[-i \int H(t) dt / \hbar]$$

2. Calculate a nontrivial case using the second relation. Carry units and uncertainty where applicable.

$$\overline{H} = \Omega_1 + \Omega_2 + \dots$$

3. Interpret the third relation in two limiting regimes and explain the physical meaning of the result.

$$P(m) = A p^m + B$$

Show enough intermediate work that another student can reproduce the calculation.